



REPORT OF FINDINGS

EDA on the Walmart Sales Dataset



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Exploratory Data Analysis on Walmart Sales Dataset

Introduction

The Walmart dataset contains historical data on weekly sales across multiple stores, including additional economic indicators like temperature, fuel prices, and unemployment rates. The purpose of this analysis is to explore how different factors affect weekly sales and uncover any patterns, trends, or anomalies that could guide better forecasting or business decisions.

Understanding the Dataset

The dataset consists of:

- Store: Store ID
- Date: Weekly record date
- Weekly_Sales: Total sales for the store that week
- Holiday_Flag: Whether the week included a major holiday
- Temperature, Fuel_Price, CPI, Unemployment: External economic/environmental metrics

Initial analysis using .info() and .describe() revealed:

- No missing values across the 6435 entries.
- Variables like Weekly_Sales and Temperature show varied distributions, which prompted further exploration.

Univariate Analysis

Weekly Sales

- Weekly sales varied significantly, with outliers during holiday weeks and promotions.
- The distribution was right-skewed, with a few weeks reaching exceptionally high sales.

Holiday Flag

- Only <10% of weeks were marked as holidays.
- These weeks generally corresponded with noticeable spikes in sales.

Temperature & Fuel Price

- Temperature ranged between 5°F and 100°F, confirming data from various regions.
- Fuel prices fluctuated between ~\$2.4 and \$4.1, indicating impact from national trends.

Bivariate Analysis

Date vs Weekly Sales

- A clear seasonal pattern was observed — sales increased sharply around holidays.
- Certain weeks showed consistent high performance across years.

Holiday Impact

- On average, holiday weeks showed a significant uplift in sales, validating the effect of promotional strategies or customer behavior.

CPI & Unemployment vs Sales

- Weak or no direct correlation was found between CPI/unemployment and sales, suggesting Walmart sales may be more resilient to macroeconomic shifts.

Multivariate Insights

Using correlation heatmaps and pairplots:

- Strongest positive correlation existed between Holiday_Flag and Weekly_Sales, as expected.
- No multicollinearity observed — each variable contributed independently.

Key Findings Summary

1. Holiday weeks drive higher sales — campaigns should align around key events.
2. Temperature and seasonality impact shopping patterns — regional trends could be considered in planning.
3. Macroeconomic indicators like CPI/unemployment had minimal effect on weekly performance, showing Walmart's operational stability.
4. Some stores consistently outperformed others, suggesting differences in location, size, or management.